<u>CAD for VLSI Design</u>

指導教授: 陳聿廣

PA1

Benchmark Translator

(Bonus)

系級 電機四 B

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1. The Answer of each questions

Q1. Total cell area = ____ (one for 130nm, another for 16nm)

- 130nm: 628.038004

```
****************************************
Report : area
Design : c432
Version: P-2019.03-SP1-1
Date   : Mon Mar  3 00:33:51 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    fast (File: /home/tools/designkit/CBDK_IC_Contest_v2.1/SynopsysDC/db/fast.db)

Number of ports:                          43
Number of nets:                          121
Number of cells:                          85
Number of combinational cells:            85
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                        20
Number of references:                     21

Combinational area:                628.038004
Buf/Inv area:                       67.895999
Noncombinational area:               0.000000
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                   628.038004
Total area:                 undefined
```

- 16nm: 22.239361

```
****************************************
Report : area
Design : c432
Version: P-2019.03-SP1-1
Date   : Mon Mar  3 00:10:22 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellss0p72vm40c_ccs (File: /home/adfp_tools/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCells
s0p72vm40c_ccs.db)

Number of ports:                          43
Number of nets:                          121
Number of cells:                          85
Number of combinational cells:            85
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                        21
Number of references:                     19

Combinational area:                22.239361
Buf/Inv area:                       3.265920
Noncombinational area:               0.000000
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   22.239361
Total area:                 undefined
```

Q2. Compare two values. Which one is smaller?

16nm。在相同的晶圓面積上，16nm 製程可以放入更多的邏輯單元，但單一 cell 的佔用面積會因 transistor 的尺寸縮小而變小。

Q3. Total power = ____ (one for 130nm, another for 16nm)

- 130nm: 9.5521 x 10^{-2} mW

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.0000	0.0000	0.0000	0.0000	(0.00%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	6.5542e-02	2.9961e-02	1.7492e+04	9.5521e-02	(100.00%)	
Total	6.5542e-02 mW	2.9961e-02 mW	1.7492e+04 pW	9.5521e-02 mW		

- 16nm: 9.4613 x 10^{-3} mW

```
                 Internal         Switching          Leakage            Total
Power Group      Power            Power              Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000            0.0000  (   0.00%)
register           0.0000           0.0000            0.0000            0.0000  (   0.00%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational  4.5293e-03       4.9317e-03            0.3448        9.4613e-03  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          4.5293e-03 mW    4.9317e-03 mW         0.3448 nW     9.4613e-03 mW
```

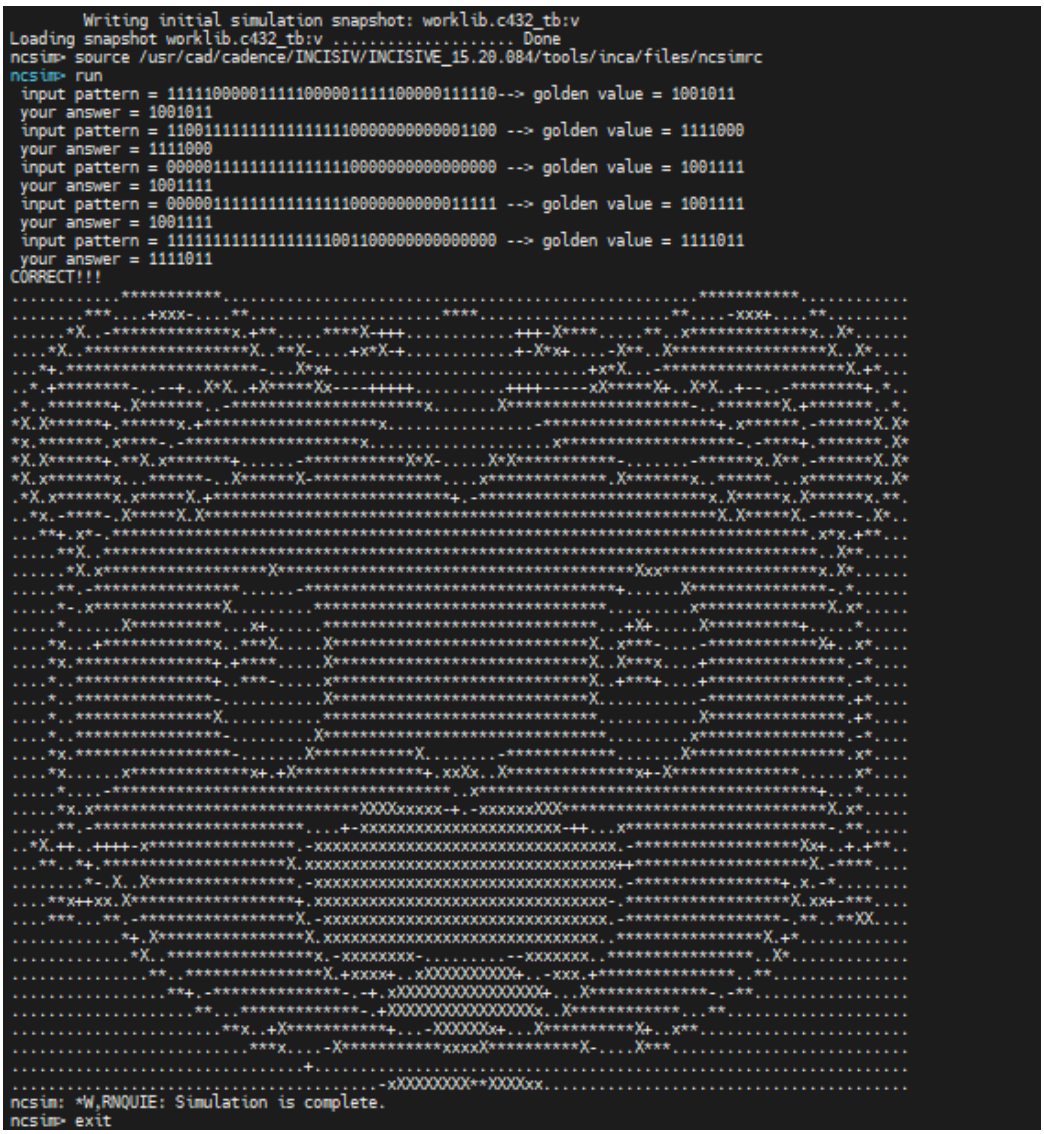
Q4. Compare two values. Which one is smaller?

16nm。由於 130nm 製程通常是使用傳統平面技術，而 16nm 製程則是使用 FinFET 技術，16nm 的三維結構能夠更好的控制閘極，且擁有更短的 channel length，大幅減少了靜態與動態功耗。

Q5. Check whether the synthesis result is correct. Have a screenshot of the result. (one for 130nm, another for 16nm)

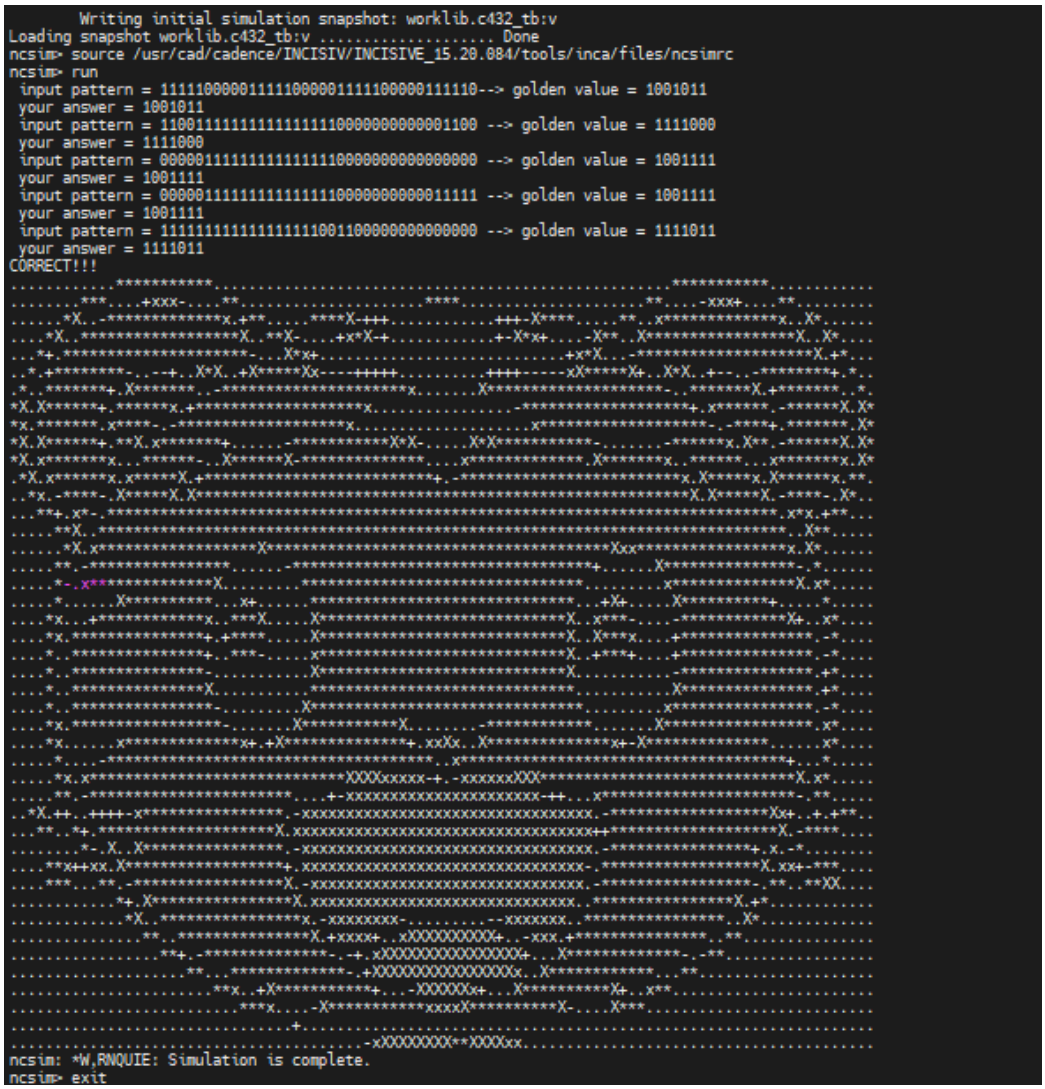
- 130nm

```
        Writing initial simulation snapshot: worklib.c432_tb:v
Loading snapshot worklib.c432_tb:v ..................... Done
ncsim> source /usr/cad/cadence/INCISIV/INCISIVE_15.20.084/tools/inca/files/ncsimrc
ncsim> run
 input pattern = 11111000001111100000111110000111110--> golden value = 1001011
 your answer = 1001011
 input pattern = 11001111111111111111000000000001100 --> golden value = 1111000
 your answer = 1111000
 input pattern = 00000111111111111111000000000000000 --> golden value = 1001111
 your answer = 1001111
 input pattern = 00000111111111111111000000000011111 --> golden value = 1001111
 your answer = 1001111
 input pattern = 11111111111111111100110000000000000 --> golden value = 1111011
 your answer = 1111011
CORRECT!!!
```



```
ncsim: *W,RNQUIE: Simulation is complete.
ncsim> exit
```

- 16nm



```
        Writing initial simulation snapshot: worklib.c432_tb:v
Loading snapshot worklib.c432_tb:v ..................... Done
ncsim> source /usr/cad/cadence/INCISIV/INCISIVE_15.20.084/tools/inca/files/ncsimrc
ncsim> run
 input pattern = 11111000000111110000011111000000111110--> golden value = 1001011
 your answer = 1001011
 input pattern = 11001111111111111111110000000000001100 --> golden value = 1111000
 your answer = 1111000
 input pattern = 00000111111111111111110000000000000000 --> golden value = 1001111
 your answer = 1001111
 input pattern = 00000111111111111111110000000000011111 --> golden value = 1001111
 your answer = 1001111
 input pattern = 11111111111111111111001100000000000000 --> golden value = 1111011
 your answer = 1111011
CORRECT!!!
ncsim: *W,RNQUIE: Simulation is complete.
ncsim> exit
```

由上面兩張圖可知，測試結果皆為正確。

2. The difficulties you encountered and how you solve the problems

在還沒添增標準延遲檔案 (SDF) 前，130nm 製程在最終驗證合成結果時會顯示失敗，且其中 testbench 某一個 case 出來的結果會有 don't care。推測是因為在邏輯合成後，產生的 gate-level netlist 沒有具體的時序資訊，導致部分訊號的時序行為不確定，因而產生 don't care。添加 SDF 能提供具體的時序資訊，使得模擬結果更能符合真實硬體行為，進而解決 don't care 的問題。

因此只要在 Design Comiler 底下添加一則指令:

```
write_sdf -version 2.1 -load_delay cell c432_130nm_syn.sdf
```

並將 sdf 檔添增至 testbench 檔生成時序標記文件(initial $sdf_annotate(`SDFFILE, uut);)，最終即能生成正確的驗證合成結果。

3. What you learn from this assignment

透過這次的作業，我學到了一些知識:

- 使用 Design Compiler 進行邏輯合成
  瞭解了設計前端流程，包括讀取 Verilog 檔案、設定製程條件、執行合成、生成 gate-level netlist 等步驟，並學會了使用命令列模式與 design vision 來執行合成流程。
- 製程技術的影響
  透過比較 16nm 與 130nm 製程的合成結果，我學會了不同製程對面積、功耗的影響。

4. Suggestions of this bonus assignment

這次的作業流程很清楚，讓我們能夠實際操作 Design Compiler 進行邏輯合成，並比較不同製程技術的影響。不過，以下是一些建議，希望能讓這次作業寫得更順暢：

- 添加指令說明
  若投影片中有一些指令操作，希望能在旁邊寫一下這個指令的內容是什麼以及為何使用這項指令(例如為何要寫 sdf 檔)，這樣能使我們更好理解這份作業的一些細節，未來能更好操作 Design Compiler 與工作站。
- 增加更多 testcase
  如果能提供更多測試案例與 testbench，會讓我們對不同電路的合成結果有更全面的理解。